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用于分子性质预测的Kolmogorov–Arnold图神经网络

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图神经网络（GNNs）作为几何深度学习中的关键模型，在分子性质预测方面表现出显著的成功。与此同时，Kolmogorov–Arnold网络（KANs）作为多层感知器的强大替代方案出现，提供了更好的表达能力、参数效率和可解释性。为了结合这两种框架的优势，我们提出了Kolmogorov–Arnold GNNs（KA-GNNs），将KAN模块整合到GNN的三个基本组成部分：节点嵌入、信息传递和读出。我们进一步在KAN中引入基于傅里叶级数的单变量函数，以增强函数逼近能力，并提供理论分析以支持其表达能力。开发了两种架构变体：KA-图卷积网络和KA-增强图注意力网络，并在七个分子基准上进行了评估。实验结果表明，KA-GNNs在预测精度和计算效率方面均持续优于传统GNNs。

即使在高效的实验工具和最先进的计算模型方面取得了巨大成功，药物设计和发现仍然是一个耗时且成本极高的过程[1]。最近，人工智能，特别是面向科学的人工智能，在科学数据分析中展示了其巨大的潜力和强大的能力。AlphaFold 模型在蛋白质结构预测方面的成功，彻底改变了分子结构与性质分析，并开启了药物设计和发现的新纪元[2–6]。一般来说，所有基于分子的人工智能模型都可分为两类，即基于分子特征的机器学习和端到端深度学习[7]。第一类依赖于分子描述符或指纹作为机器学习模型的输入特征。关键过程是分子特征化（或特征工程），它从结构、物理、化学或被证明非常有效[8–11]。将这些基于拓扑的分子特征与机器学习模型结合，在药物设计的各个阶段取得了显著成功，如蛋白质-配体结合亲和力预测[12,13]、、蛋白质突变分析[14,15] 等[16,17]。

第二类包括端到端的深度学习模型，这些模型使用各种分子表示方法，例如简化分子输入线表示法（SMILES）字符串、基于分子的图像和体积数据，或分子图，并采用诸如Transformer、二维（2D）或三维（3D）卷积神经网络（CNN）以及几何深度学习（GDL）等架构。特别是，GDL模型，如图卷积网络（GCN）、图自编码器、图Transformer等，已被广泛应用于分子数据分析和药物设计。然而，传统基于共价键的分子图表示存在各种局限性，而引入非共价相互作用已被证明显著提升性能。

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Kolmogorov–Arnold graph neural networks for molecular property prediction

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Graph neural networks (GNNs) have shown remarkable success in molecular property prediction as key models in geometric deep learning. Meanwhile, Kolmogorov–Arnold networks (KANs) have emerged as powerful alternatives to multi-layer perceptrons, offering improved expressivity, parameter efficiency and interpretability. To combine the strengths of both frameworks, we propose Kolmogorov–Arnold GNNs (KA-GNNs), which integrate KAN modules into the three fundamental components of GNNs: node embedding, message passing and readout. We further introduce Fourier-series-based univariate functions within KAN to enhance function approximation and provide theoretical analysis to support their expressiveness. Two architectural variants, KA-graph convolutional networks and KA-augmented graph attention networks, are developed and evaluated across seven molecular benchmarks. Experimental results show that KA-GNNs consistently outperform conventional GNNs in terms of both prediction accuracy and computational efficiency. Moreover, our models exhibit improved interpretability by highlighting chemically meaningful substructures. These findings demonstrate that KA-GNNs offer a powerful and generalizable framework for molecular data modelling, drug discovery and beyond.

Even with the huge successes in efficient experimental tools and state-of-the-art computational models, drug design and discovery is still a time-consuming and extremely costly process[1]. Recently, artificial intelligence, in particular artificial intelligence for sciences, has demonstrated its enormous potential and great power in scientific data analysis. The success of AlphaFold models in protein structure prediction has fundamentally changed molecular structural and property analysis, and ushered in a new era of drug design and discovery[2–6]. In general, all molecule-based artificial intelligence models fall into two categories, that is, molecular feature-based machine learning and end-to-end deep learning[7]. The first category relies on molecular descriptors or fingerprints as input features for machine learning models. The key process is molecular featurization (or feature engineering), which extracts or generates molecular features from structural, physical, chemical or biological properties. Among them, structure-based descriptors or fingerprints, especially those derived from topological methods, have proved highly effective[8–11]. Integrating these topology-based molecular features with machine learning models has led to notable success in various drug design stages, such as protein–ligand binding affinity prediction[12,13], protein mutation analysis[14,15] and others[16,17].

The second category includes end-to-end deep learning models that use various molecular representations, such as simplified molecular input line entry system strings, molecule-based images and volumetric data, or molecular graphs, and adopt architectures such as Transformers, two-dimensional (2D) or three-dimensional (3D) convolutional neural networks (CNNs) and geometric deep learning (GDL)[18–23]. In particular, GDL models, such as graph convolutional networks (GCNs)[24], graph autoencoders[25], graph transformers[26] and so on, have been widely used in molecular data analysis and drug design. However, traditional covalent-bond-based molecular graph representations have various limitations, while incorporating non-covalent interactions has been shown to notably enhance performance[27]. These

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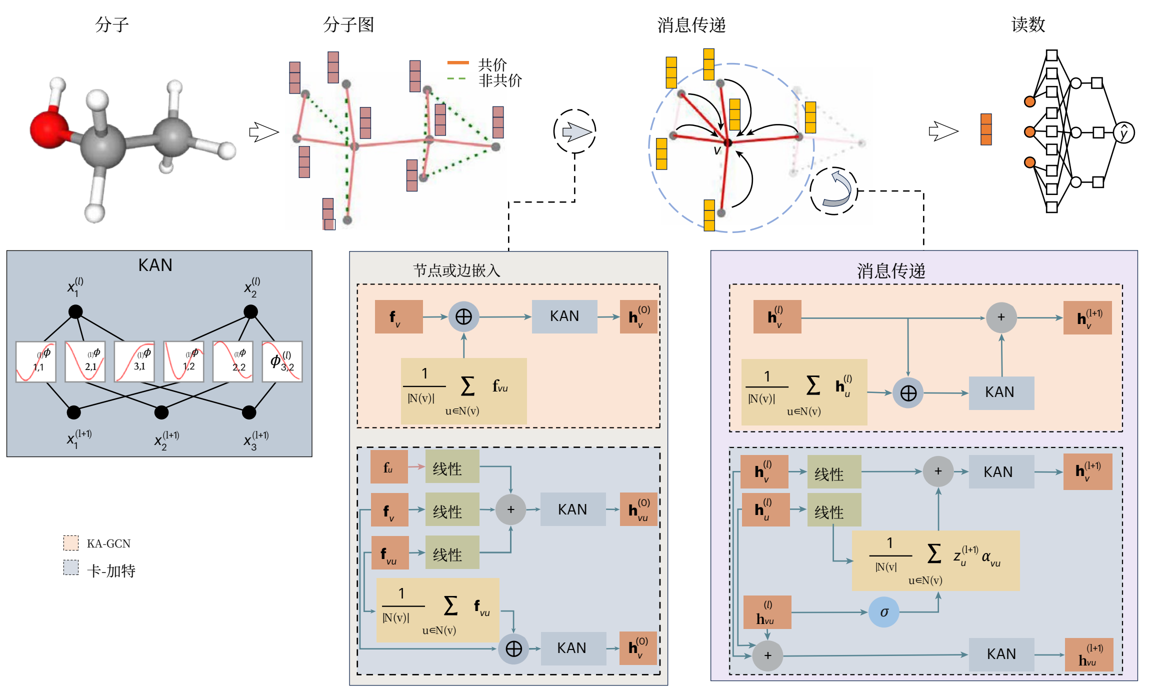

图 1 | KA-GNN 模型架构概述 结构。流程图展示了GNN中修改过的组件：节点或边嵌入、信息传递 ，池化和预测模块。

结果表明，新的图表示可能优于传统的基于共价键的表示。通过将基于几何和/或拓扑的分子图整合到GDL模型中，我们可以提升模型性能，并深入了解分子结构和功能。

Kolmogorov–Arnold 网络（KANs）基于 Kolmogorov–Arnold 表示定理，正在成为传统多层感知机（MLPs）的一个有吸引力的替代方案。与在边上使用常数权重、在节点上使用激活函数的传统 MLP 不同，KAN 在边上采用可学习的（一元）基函数，从而实现复杂函数的精确且可解释的建模。它们不仅在求解偏微分方程方面显示出潜力[29]，，还在各种应用领域表现出前景[30–33]。例如，一项研究[33]通过将 KAN 与长短期记忆网络结合，实现了多步时间序列预测的显著改进。另一项研究[34]通过将 KAN 融入预训练卷积神经网络，增强了遥感场景分类。对一元函数的选择也极大地影响性能。之前的一项研究[35]表明，小波函数可提高可解释性和频率分析。同样，另一项研究

更重要的是，KAN 模块已被嵌入到图神经网络（GNN）中，以取代节点嵌入、消息传递和读出中使用的 MLP。例如，GraphKAN[37] 在嵌入和读出部分应用了 KAN。GKAN[38] 和 GraphKAN[39] 在消息传递过程中使用了 B 样条函数。KA-GNNs[30] 和 GNN-SKAN[40] 在消息传递中使用了径向基函数，其中 GNN-SKAN 还改进了读出。所有这些方法都增强了单个 GNN 组件，并且优于它们的原始模型，展示了 KAN 集成的强大功能。

在这里，我们提出了KA-GNN，一个统一的框架，能够将KANs充分整合到GNN的三个核心组件中，包括节点嵌入、信息传递和读出。基于傅里叶的一元函数被开发用于有效捕捉图中的低频和高频结构模式，从而增强特征嵌入和信息聚合的表达能力。一个理论

提供了证明以展示我们 Fourier-KAN 模型的强近似能力。我们进一步设计了两个变体：KA-GCN（KAN 增强的 GCN）和 KAN 增强的图注意力网络（KA-GAT），如图 1 所示。在七个基准数据集上的广泛实验验证了 KA-GNN 的卓越准确性和计算效率，确立了其作为非欧几里得数据 GDL 中有前途的新范式的地位。

结果

基于傅里叶的KAN层

KANs 的灵感来自 Kolmogorov–Arnold 叠加定理，该定理指出任何多变量连续函数都可以表示为有限个单变量函数和加法的组合。因此，KANs 通过将每一层构建为可学习的单变量函数之和来对其进行泛化，这与叠加形式一致。每个函数充当可学习的预激活，为 MLP 中固定非线性（例如 ReLU）提供了一种灵活的替代方案。这使得 KANs 能够使用更少的参数并表现出更平滑的梯度，从而产生更紧凑且更精确的函数近似。

在这项工作中，我们通过采用傅里叶级数作为其预激活函数的基函数（方程（4）和（5））来扩展KANs。与之前使用B样条函数的KAN实现相比，基于傅里叶的公式能够有效地捕捉图中的低频和高频结构模式。使用全局三角函数可以实现平滑、紧凑的表示，从而有利于梯度流动和参数效率。

我们从理论上证明了所提出的基于傅里叶的 KANs 架构具有强大的逼近能力。尽管该结构的灵感来源于 Kolmogorov–Arnold 表示定理，但该定理中涉及的单变量函数可能高度不平滑，这限制了它作为理论基础的适用性。为了严格奠定我们模型的表达能力，我们改为将理论基础建立在 Carleson 的收敛定理和 Fefferman 的多变量扩展之上[41,42]：

定理 1. 设 Z^n 表示 n 维整数格，且 $Z^n_N = \{1, 2, \ldots, N\}^n \subset Z^n$。对于平方可积函数 $f \in L^2([0, 2\pi]^n)$ 及其傅里叶展开式：

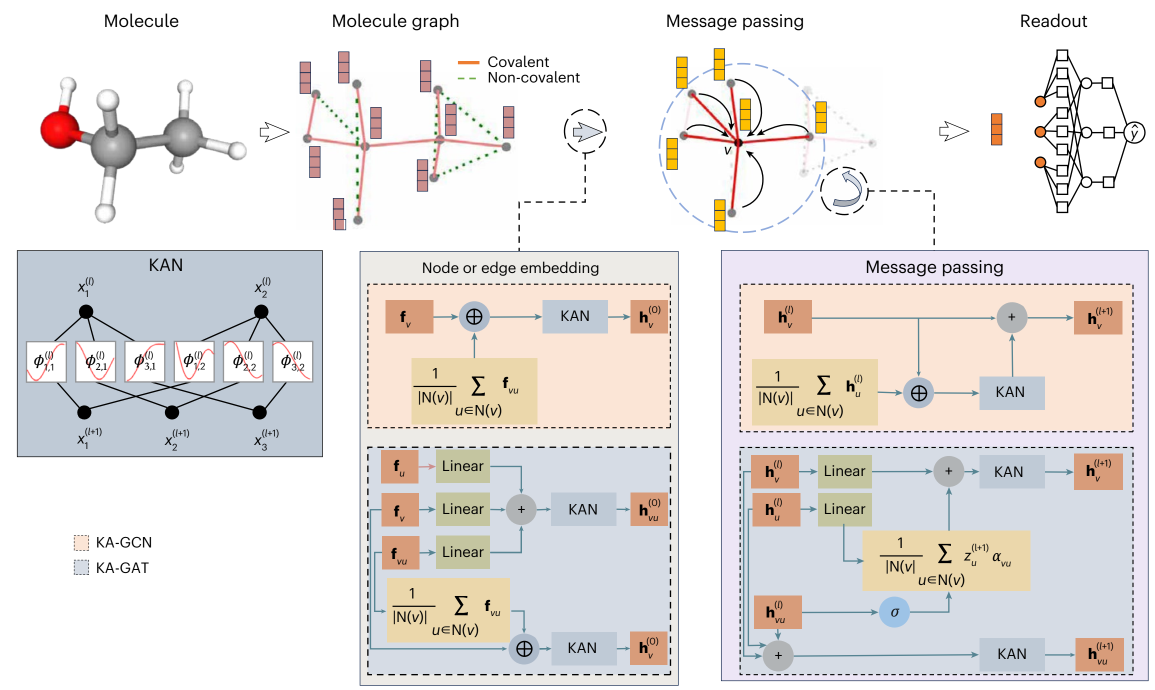

Fig. 1 | Overview of the KA-GNN model architecture. The flowchart illustrates the modified components within the GNN: node or edge embedding, message passing, pooling and prediction modules.

results indicate that new graph representations could outperform traditional covalent-bond-based ones. By integrating geometry and/or topology-based molecular graphs into GDL models, we can boost model performance and gain deeper insights into molecular structure and function[28].

Kolmogorov–Arnold networks (KANs), grounded in the Kolmogorov–Arnold representation theorem, are emerging as a compelling alternative to traditional multi-layer perceptrons (MLPs). Unlike conventional MLPs that use constant weights on edges and activation functions on nodes, KANs adopt learnable (basis) univariate functions on edges, enabling accurate and interpretable modelling of complex functions. They have shown promise not only in solving partial differential equations[29], but also across diverse application domains[30–33]. For instance, one study[33] achieved notable improvements in multi-step time series forecasting by combining KANs with long short-term memory networks. Another study[34] enhanced remote sensing scene classification by incorporating KANs into pretrained CNNs. The choice of univariate functions also greatly influences performance. A previous study[35] demonstrated that wavelet functions improve interpretability and frequency analysis. Similarly, another study[36] used Jacobi polynomials in their PointNet-KAN model to achieve strong results with a shallower, more efficient architecture.

More importantly, KAN modules have been embedded into graph neural networks (GNNs) to replace the MLPs used in node embedding, message passing and readout. For instance, GraphKAN[37] applied KANs in the embedding and readout parts. GKAN[38] and GraphKAN[39] used B-spline functions during message passing. KA-GNNs[30] and GNN-SKAN[40] used radial basis functions for message passing, with GNN-SKAN also improving the readout. All these approaches enhanced individual GNN components and outperformed their original models, demonstrating the power of KAN integration.

Here we propose KA-GNN, a unified framework that fully integrates KANs into all three core components of GNNs, including node embedding, message passing and readout. Fourier-based univariate functions are developed to effectively capture both low-frequency and high-frequency structural patterns in graphs, enhancing the expressiveness of feature embedding and message aggregation. A theoretical proof is provided to demonstrate the strong approximation capabilities of our Fourier-KAN model. We further design two variants: KA-GCN (KAN-augmented GCN) and KAN-augmented graph attention network (KA-GAT), as shown in Fig. 1. Extensive experiments across seven benchmark datasets validate KA-GNN's superior accuracy and computational efficiency, establishing it as a promising new paradigm in GDL for non-Euclidean data.

Results

Fourier-based KAN layer

KANs are inspired by the Kolmogorov–Arnold superposition theorem, which states that any multivariate continuous function can be expressed as a finite composition of univariate functions and addition. Accordingly, KANs generalize this by constructing each layer as a sum of learnable univariate functions, consistent with the superposition form. Each function acts as a learnable pre-activation, offering a flexible alternative to fixed nonlinearities (for example, ReLU) in MLPs. This enables KANs to use fewer parameters and exhibit smoother gradients, yielding more compact and accurate function approximations.

In this work, we extend the KANs by adopting the Fourier series as the basis for its pre-activation functions (equations (4) and (5)). Compared with previous KAN implementations that use B-spline functions, the Fourier-based formulation enables both low-frequency and high-frequency structural patterns in graphs to be effectively captured. The use of global trigonometric functions allows for smooth, compact representations that benefit gradient flow and parameter efficiency.

We theoretically establish that the proposed Fourier-based KANs architecture possesses strong approximation capability. Although this structure is inspired by the Kolmogorov–Arnold representation theorem, the univariate functions involved in that theorem can be highly non-smooth, limiting its suitability as a theoretical foundation. To rigorously ground the expressive power of our model, we instead base our theoretical foundation on Carleson's convergence theorem and Fefferman's multivariate extension[41,42]:

Theorem 1. *Let Z^n denote the n-dimensional integer lattice, and $Z^n_N = \{1, 2, \ldots, N\}^n \subset Z^n$. For a square-integrable function $f \in L^2([0, 2\pi]^n)$ and its Fourier expansion:*

$$f(\mathbf{x}) \sim \sum_{\mathbf{k}\in Z^n} (a_{\mathbf{k}}\cos(\mathbf{k}\cdot\mathbf{x}) + b_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{x})),$$

当 $x \in [0, 2\pi]^n$ 时，我们有

$$f(\mathbf{x}) = \lim_{N\to\infty} \sum_{\mathbf{k}\in Z_N^n} (a_{\mathbf{k}}\cos(\mathbf{k}\cdot\mathbf{x}) + b_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{x}))$$

几乎到处

受到傅里叶级数强近似保证的激励，我们将它们作为模型的基础组成部分。我们保留 KAN 结构，但将预激活替换为傅里叶级数。以下结果进一步支持了其表达能力：

定理 2. 设 $f \in L^2([0, 2\pi]^n)$ 是一个平方可积函数。对于几乎所有 $x \in [0, 2\pi]^n$ 和任意 $\epsilon > 0$，存在一个正整数 K 以及一列基于傅里叶的 KAN：KANₗ，对于 $l = 0, 1, \ldots, L$，使得预激活函数中的谐波数量被 K 所限制，并且

$$|f(\mathbf{x}) - \mathrm{KAN}_L \circ \mathrm{KAN}_{L-1} \circ \cdots \circ \mathrm{KAN}_0(\mathbf{x})| < \epsilon,$$

其中 ∘ 表示函数的复合。

这些理论结果表明，基于傅里叶的 KAN 架构可以逼近任意平方可积的多变量函数，提供了强大的表达能力和理论保证。为了在实证上验证这一点，我们将其拟合性能与标准的两层 MLP 在六个代表性函数上的表现进行比较。补充图 1 的结果展示了我们方法的优越逼近能力，证实了将傅里叶级数纳入 KAN 框架的实际益处。

KA-GNNs

受到基于傅里叶的KAN层理论优势的启发，我们开发了一类GNNs，称为KA-GNNs，它系统地将基于傅里叶的KAN模块集成到整个GNN流程中，包括节点嵌入初始化、消息传递和图级读取。通过这种集成，用基于傅里叶的KAN模块取代传统的基于MLP的变换，从而形成一个统一的、可完全微分的架构，并具有增强的表示能力和改进的训练动态。通过利用自适应、数据驱动的非线性映射，而不是固定激活函数，KA-GNNs构建了更丰富的节点嵌入，在消息传递过程中调节特征交互，并捕获更具表达力的图级表示。

为了实例化 KA-GNNs 框架，我们设计了两种变体，即 KA-GCN 和 KA-GAT，它们分别将基于傅里叶的 KAN 模块整合到 GCN 和 GAT 主干中。在 KA-GCN 中，每个节点的初始嵌入是通过将其原子特征（例如，原子序数、半径）与其相邻键特征的平均值（例如，键类型、键长）连接后传入 KAN 层计算得到的。这一设计通过数据依赖的三角变换编码了原子身份和局部化学环境。消息传递层遵循 GCN 方案，而节点特征通过残差 KAN 更新，从而替代了传统的 MLP。

KA-GAT 通过结合边嵌入引入了一种富于表现力的设计。节点和边的特征均使用 KAN 层进行初始化。每条边的嵌入是通过将其键特征与两个端点原子的特征融合形成的，从而实现上下文感知的键表示。在消息传递过程中，注意力分数由边嵌入计算，并且节点和边特征通过 KAN 层迭代更新。这支持在共价键和非共价键之间进行自适应注意力和丰富的特征交互。通过用 KAN 驱动的映射替代静态聚合和固定的 MLP，KA-GCN 和 KA-GAT 实现了更灵活和富有表现力的分子图建模。

用于分子性质预测的KA-GNNs

为了将 KA-GCN 和 KA-GAT 应用于分子数据，我们将每个分子表示为图 G(V, E)，其中原子为节点 (V)，共价和非共价相互作用为边 (E)。具体来说，如果两个原子之间存在键连接或在 5 Å 截断距离内空间上接近，则在它们之间建立一条边，从而捕捉局部键和长程依赖。每个原子使用一个 92 维向量进行编码，遵循晶体图卷积神经网络的方法，包含原子序数、主族、周期、电负性和共价半径。每个键使用一个 21 维向量进行编码：共价键使用化学描述符表示，而非共价键使用空间描述符表示。这两种键类型被视为不同的边类别，并采用不同的初始编码设置。这种丰富的图表示为 KA-GCN 和 KA-GAT 提供了化学和空间上下文，使基于傅里叶的 KAN 层能够有效地建模复杂的原子间关系。

KA-GNNs的性能与比较

为了评估我们提出的 KA-GNN 模型的性能，我们考虑了来自 MoleculeNet[44] 的七个广泛使用的基准数据集。其中，三个数据集，包括 MUV、HIV 和 BACE，来自生物物理学领域，而其余四个包括 BBBP、Tox21、SIDER 和 ClinTox，则属于生理学领域。

在我们对七个数据集的比较分析中，我们评估了一系列GDL模型，包括AttentiveFP[45]、D-MPNN[46]、Mol-GDL[28]、N-Gram[47]、PretrainGNN[48]、GraphMVP[49]、MolCLR[50]、GEM[51]、Uni-mol[52]和SMPT[53]。除了我们的KA-GNN模型外，我们还包括其他基于KAN的GNN架构，如GNN-SKAN[40]、GraphKAN[39] 和KA-GNNs[30]。

为了确保公平比较，我们在所有模型中采用一致的实验设置。超参数配置和输入特征构建的详细信息见补充表1和表2。

表 1 的比较结果进一步证实了 KA-GNN 模型的优越性。我们的模型在所有基准数据集上都取得了最先进的性能，尤其是在复杂且具有挑战性的数据集如 ClinTox 和 MUV 上表现出色。这些结果表明了我们的模型在处理分子数据方面的强大能力。值得注意的是，在 BBBP 数据集中，KA-GCN 和 KA-GAT 的曲线下面积（AUC）分别提高了约 7.95% 和 7.68%。

傅里叶级数对于KA-GNN的重要性

我们在 GNN 架构中使用基于傅里叶的 KAN 作为核心模块，区别于传统的基于 MLP 的设计。为了评估基函数对分子性质预测的影响，我们在 KA-GCN 和 KA-GAT 框架内评估了 B 样条、多项式和傅里叶级数的形式。所有变体都是在相同的 GCN 和 GAT 设置下进行基准测试的。多项式变换在方程 (1) 中定义：

$$x_j^{(l+1)} = \sum_{i=1}^{n_l}\sum_{k=0}^{K}\left(C_{k,j,i}^{(l)}\left(x_i^{(l)}\right)^k\right), \quad (1)$$

其中 $C^{(l)}_{k,j,i}$ 表示可学习参数，初始化自 $\mathcal{N}(0, \frac{1}{n_{l+1}\times K})$，而 $\mathbf{x}^{(l)} = (x_1^{(l)}, x_2^{(l)}, \cdots, x_{n_l}^{(l)})$ 是第 l 层的激活向量。

表2显示了跨数据集的一致性能提升，验证了KA-GNN在分子任务中的有效性。基于傅里叶的KAN的卓越表达能力支撑了这些结果，这在定理2中得到了理论支持，并在我们的函数拟合实验中得到了实证验证。此外，我们的消融研究表明，基于傅里叶的KAN不仅提高了准确性，还改善了特征嵌入、消息传递和最终预测阶段的表现。

我们进一步将标准的 GCN/GAT 模型与它们的 KA-GNN 对应模型（使用傅里叶级数）在相同的分子图下进行比较。如表 3 所示，KA-GNN 始终优于基准模型，

$$f(\mathbf{x}) \sim \sum_{\mathbf{k}\in Z^n} (a_{\mathbf{k}}\cos(\mathbf{k}\cdot\mathbf{x}) + b_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{x})),$$

where $x \in [0, 2\pi]^n$, *we have*

$$f(\mathbf{x}) = \lim_{N\to\infty} \sum_{\mathbf{k}\in Z_N^n} (a_{\mathbf{k}}\cos(\mathbf{k}\cdot\mathbf{x}) + b_{\mathbf{k}}\sin(\mathbf{k}\cdot\mathbf{x}))$$

almost everywhere.

Motivated by the strong approximation guarantees of Fourier series, we adopt them as foundational components of our model. We retain the KAN structure but replace pre-activations with Fourier series. The following result further supports its expressive power:

Theorem 2. *Let* $f \in L^2([0, 2\pi]^n)$ *be a square-integrable function. For almost every* $\mathbf{x} \in [0, 2\pi]^n$ *and any* $\epsilon > 0$*, there exists a positive integer K and a sequence of Fourier-based* KANs: KAN_l*, for* $l = 0, 1, \ldots, L$*, such that the number of harmonics in pre-activation functions is bounded by K, and*

$$|f(\mathbf{x}) - \mathrm{KAN}_L \circ \mathrm{KAN}_{L-1} \circ \cdots \circ \mathrm{KAN}_0(\mathbf{x})| < \epsilon,$$

where ∘ *denotes function composition.*

These theoretical results show that the Fourier-based KAN architecture can approximate any square-integrable multivariate function, providing strong expressive power and theoretical guarantees. To empirically validate this, we compare its fitting performance with a standard two-layer MLP across six representative functions. Results in Supplementary Fig. 1 demonstrate the superior approximation capability of our approach, confirming the practical benefits of incorporating Fourier series into the KAN framework.

KA-GNNs

Inspired by the theoretical strengths of the Fourier-based KAN layer, we develop a class of GNNs, known as KA-GNNs, that systematically integrate Fourier-based KAN modules across the entire GNN pipeline, including node embedding initialization, message passing and graph-level readout. This integration replaces conventional MLP-based transformations with Fourier-based KAN modules, yielding a unified, fully differentiable architecture with enhanced representational power and improved training dynamics. By leveraging adaptive, data-driven nonlinear mappings over fixed activations, KA-GNNs construct richer node embeddings, modulate feature interactions during message passing and capture more expressive graph-level representations.

To instantiate the KA-GNNs framework, we design two variants, that is, KA-GCN and KA-GAT, which respectively integrate Fourier-based KAN modules into GCN and GAT backbones. In KA-GCN, each node's initial embedding is computed by passing the concatenation of its atomic features (for example, atomic number, radius) and the average of its neighbouring bond features (for example, bond type, bond length) through a KAN layer. This design encodes both atomic identity and local chemical context via data-dependent trigonometric transformations. The message-passing layers follow the GCN scheme, while node features are updated via residual KANs, replacing traditional MLPs.

KA-GAT introduces an expressive design by also incorporating edge embeddings. Both node and edge features are initialized using KAN layers. Each edge embedding is formed by fusing its bond features with features of the two endpoint atoms, enabling context-aware bond representations. During message passing, attention scores are computed from edge embeddings, and both node and edge features are iteratively updated via KAN layers. This supports adaptive attention and rich feature interactions across covalent and non-covalent bonds. By replacing static aggregation and fixed MLPs with KAN-driven mappings, KA-GCN and KA-GAT achieve more flexible and expressive molecular graph modelling.

KA-GNNs for molecular property prediction

To apply KA-GCN and KA-GAT to molecular data, we represent each molecule as a graph $G = (V, E)$, where atoms are nodes (V) and both covalent and non-covalent interactions are edges (E). Specifically, an edge connects two atoms if they are bonded or spatially close within a 5-Å cut-off, capturing both local bonds and long-range dependencies. Each atom is encoded with a 92-dimensional vector following crystal graph convolutional neural networks[43], incorporating atomic number, group, period, electronegativity and covalent radius. Each bond is encoded with a 21-dimensional vector: covalent bonds are represented using chemical descriptors, while non-covalent bonds use spatial descriptors. These two bond types are treated as distinct edge categories with different initial encoding settings. This enriched graph representation equips KA-GCN and KA-GAT with both chemical and spatial context, allowing Fourier-based KAN layers to effectively model complex interatomic relationships.

Performance and comparison of KA-GNNs

To evaluate the performance of our proposed KA-GNN models, we consider seven widely used benchmark datasets from MoleculeNet[44]. Of these, three datasets, including MUV, HIV and BACE, are from the biophysics domain, while the remaining four including BBBP, Tox21, SIDER and ClinTox, belong to the physiology domain.

In our comparative analysis across seven datasets, we evaluated a range of GDL models, including AttentiveFP[45], D-MPNN[46], Mol-GDL[28], N-Gram[47], PretrainGNN[48], GraphMVP[49], MolCLR[50], GEM[51], Uni-mol[52] and SMPT[53]. In addition to our KA-GNN models, we also include other KAN-based GNN architectures, such as GNN-SKAN[40], GraphKAN[39] and KA-GNNs[30].

To ensure fair comparisons, we adopt consistent experimental settings across all models. Details of the hyperparameter configurations and input feature constructions are provided in Supplementary Tables 1 and 2.

The comparative results in Table 1 further confirm the superiority of the KA-GNN models. Our model achieves state-of-the-art performance across all benchmark datasets, particularly excelling on complex and challenging datasets such as ClinTox and MUV. These results demonstrate the robust capability of our model in handling molecular data. Notably, in the BBBP dataset, the area under the curve (AUC) for KA-GCN and KA-GAT showed improvements of approximately 7.95% and 7.68%.

Importance of Fourier series for KA-GNNs

We use a Fourier-based KAN as the core module in our GNN architecture, departing from traditional MLP-based designs. To assess the influence of basis functions on molecular property prediction, we evaluate B-spline, polynomial and Fourier-series formulations within both KA-GCN and KA-GAT frameworks. All variants are benchmarked under identical GCN and GAT settings. The polynomial transformation is defined in equation (1):

$$x_j^{(l+1)} = \sum_{i=1}^{n_l}\sum_{k=0}^{K}\left(C_{k,j,i}^{(l)}\left(x_i^{(l)}\right)^k\right), \quad (1)$$

where $C^{(l)}_{k,j,i}$ denotes learnable parameters initialized from $\mathcal{N}(0, \frac{1}{n_{l+1}\times K})$, and $\mathbf{x}^{(l)} = (x_1^{(l)}, x_2^{(l)}, \ldots, x_{n_l}^{(l)})$ is the activation vector at layer l.

Table 2 shows consistent performance gains across datasets, validating the effectiveness of KA-GNNs for molecular tasks. The superior expressiveness of Fourier-based KANs underpins these results, supported theoretically in Theorem 2 and empirically in our function-fitting experiments. Further, our ablation study reveals that Fourier-based KANs improve not only accuracy but also feature embedding, message passing and final prediction stages.

We further compare standard GCN/GAT models with their KA-GNN counterparts (using Fourier series) under the same molecular graphs. As shown in Table 3, KA-GNNs consistently outperform baselines,

表 1 | KA-GN 比较　　在各种分子性质预测数据集中，使用最先进的GDL模型的Ns　　ts

模型	BACE	BBBP	临床毒理	侧面	Tox21	HIV	MUV
分子数量	1,513	2,039	1,478	1,427	7,831	41,127	93,808
平均原子数	65	46	50.58	65	36	46	43
任务数量	1	1	2	27	12	1	17
D-MPNN[46]	$0.809_{(0.006)}$	$0.710_{(0.003)}$	$0.906_{(0.007)}$	$0.570_{(0.007)}$	$0.759_{(0.007)}$	$0.771_{(0.005)}$	$0.786_{(0.014)}$
AttentiveFP[45]	$0.784_{(0.022)}$	$0.663_{(0.018)}$	$0.847_{(0.003)}$	$0.606_{(0.032)}$	$0.781_{(0.005)}$	$0.757_{(0.014)}$	$0.786_{(0.015)}$
N-GramRF[47]	$0.779_{(0.015)}$	$0.697_{(0.006)}$	$0.775_{(0.040)}$	$0.668_{(0.007)}$	$0.743_{(0.009)}$	$0.772_{(0.004)}$	$0.769_{(0.002)}$
N-GramXGB[47]	$0.791_{(0.013)}$	$0.691_{(0.008)}$	$0.875_{(0.027)}$	$0.655_{(0.007)}$	$0.758_{(0.009)}$	$0.787_{(0.004)}$	$0.748_{(0.002)}$
预训练GNN[48]	$0.845_{(0.007)}$	$0.687_{(0.013)}$	$0.726_{(0.015)}$	$0.627_{(0.008)}$	$0.781_{(0.006)}$	$0.799_{(0.007)}$	$0.813_{(0.021)}$
GraphMVP[49]	$0.812_{(0.009)}$	$0.724_{(0.016)}$	$0.791_{(0.028)}$	$0.639_{(0.012)}$	$0.759_{(0.005)}$	$0.770_{(0.012)}$	$0.777_{(0.006)}$
MolCLR[50]	$0.824_{(0.009)}$	$0.722_{(0.021)}$	$0.912_{(0.035)}$	$0.589_{(0.014)}$	$0.750_{(0.002)}$	$0.781_{(0.005)}$	$0.796_{(0.019)}$
GEM[51]	$0.856_{(0.011)}$	$0.724_{(0.004)}$	$0.901_{(0.013)}$	$0.672_{(0.004)}$	$0.781_{(0.001)}$	$0.806_{(0.009)}$	$0.817_{(0.005)}$
Mol-GDL[28]	$0.863_{(0.019)}$	$0.728_{(0.019)}$	$0.966_{(0.002)}$	$0.831_{(0.002)}$	$0.794_{(0.005)}$	$0.808_{(0.007)}$	$0.675_{(0.014)}$
Uni-mol[52]	$0.857_{(0.002)}$	$0.729_{(0.006)}$	$0.919_{(0.018)}$	$0.659_{(0.013)}$	$0.796_{(0.005)}$	$0.808_{(0.003)}$	$0.821_{(0.013)}$
SMPT[53]	$0.873_{(0.015)}$	$0.734_{(0.003)}$	$0.927_{(0.002)}$	$0.676_{(0.050)}$	$0.797_{(0.001)}$	$0.812_{(0.001)}$	$0.822_{(0.008)}$
GNN-SKAN[40]	$0.747_{(0.009)}$	$0.676_{(0.014)}$	-	$0.614_{(0.005)}$	$0.747_{(0.005)}$	$0.786_{(0.015)}$	-
GraphKAN[39]	$0.823_{(0.011)}$	$0.731_{(0.017)}$	$0.984_{(0.003)}$	$0.837_{(0.001)}$	$0.753_{(0.007)}$	$0.711_{(0.016)}$	$0.715_{(0.014)}$
KA-GNNs[30]	$0.752_{(0.011)}$	$0.721_{(0.003)}$	$0.972_{(0.001)}$	$0.831_{(0.004)}$	$0.730_{(0.012)}$	$0.717_{(0.018)}$	$0.701_{(0.006)}$
KA-GCN	**$0.890_{(0.014)}$**	**$0.787_{(0.014)}$**	**$0.992_{(0.005)}$**	$0.842_{(0.001)}$	$0.799_{(0.005)}$	$0.821_{(0.005)}$	**$0.834_{(0.009)}$**
卡-加特	$0.884_{(0.004)}$	$0.785_{(0.021)}$	$0.991_{(0.005)}$	**$0.847_{(0.002)}$**	$0.800_{(0.006)}$	**$0.823_{(0.002)}$**	$0.834_{(0.010)}$

注意，粗体数值表示性能最好的模型；下划线数值表示第二好的模型。所有指标都是平均受试者工作特征曲线下面积（AUC），标准差以下标形式显示。

表 2 | KA-GNN（KA-GCN 和 KA-GAT）模型在三种类型基函数下的性能比较，包括 B 样条、多项式和傅里叶级数

数据集	KA-GCN 模型			KA-GAT 模型		
	B样条	多项式	傅里叶级数	B样条	多项式	傅里叶级数
BACE	$0.771_{(0.012)}$	$0.853_{(0.027)}$	**$0.890_{(0.014)}$**	$0.808_{(0.009)}$	$0.8319_{(0.007)}$	**$0.884_{(0.004)}$**
BBBP	$0.723_{(0.008)}$	$0.654_{(0.009)}$	**$0.787_{(0.014)}$**	$0.657_{(0.004)}$	$0.708_{(0.005)}$	**$0.785_{(0.021)}$**
临床毒理	$0.973_{(0.006)}$	$0.981_{(0.004)}$	**$0.992_{(0.005)}$**	$0.948_{(0.003)}$	$0.983_{(0.003)}$	**$0.991_{(0.005)}$**
侧面	$0.824_{(0.003)}$	$0.832_{(0.006)}$	**$0.842_{(0.001)}$**	$0.825_{(0.004)}$	$0.836_{(0.001)}$	**$0.847_{(0.002)}$**
Tox21	$0.724_{(0.005)}$	$0.715_{(0.004)}$	**$0.799_{(0.005)}$**	$0.731_{(0.012)}$	$0.753_{(0.004)}$	**$0.800_{(0.006)}$**
HIV	$0.753_{(0.007)}$	$0.804_{(0.010)}$	**$0.821_{(0.005)}$**	$0.744_{(0.018)}$	$0.818_{(0.006)}$	**$0.823_{(0.002)}$**
MUV	$0.638_{(0.008)}$	$0.787_{(0.012)}$	**$0.834_{(0.009)}$**	$0.792_{(0.013)}$	$0.797_{(0.011)}$	**$0.834_{(0.010)}$**

请注意，加粗的数值表示　　是表现最好的模型。所有指标都是平均接收器工作特性曲线下的面积（AUC）。标准差用下标表示　　s.

突出傅里叶KANs在捕捉化学结构表示方面的增强能力。

为了研究 KA-GNNs 性能背后的架构驱动因素，我们进行了广泛的消融研究，考察了输入归一化、KAN 模块化、图构建（例如，非共价键）、回归能力、可扩展性和超参数敏感性（补充表 3–8）。

值得注意的是，我们测试了六种节点和边特征的归一化方案。KA-GNN在所有方案中都保持了强劲的性能，凸显了其对输入尺度变化的鲁棒性。

基于傅里叶的KAN在图神经网络中的效率

如图2a、b所示，基于傅里叶的KAN模型始终比B样条KAN以及基于MLP的GCN/GAT具有更低的运行时间。这种高效性源于正弦基函数的全局性质，它们能够以更少的可训练参数实现紧凑的函数近似。

与具有大型密集权重矩阵的多层感知器（MLP）不同，KAN使用固定的基函数和轻量级可学习系数，从而显著减少

表 3 | GCN/GAT 模型与 KA-GCN/KA-GAT 的傅里叶级数比较

数据集	GCN	KA-GCN	GAT	卡-加特
BACE	$0.835_{(0.014)}$	**$0.890_{(0.014)}$**	$0.834_{(0.012)}$	**$0.884_{(0.004)}$**
BBBP	$0.735_{(0.011)}$	**$0.787_{(0.014)}$**	$0.707_{(0.007)}$	**$0.785_{(0.021)}$**
临床毒理	$0.979_{(0.004)}$	**$0.992_{(0.005)}$**	$0.983_{(0.006)}$	**$0.991_{(0.005)}$**
侧面	$0.834_{(0.001)}$	**$0.842_{(0.001)}$**	$0.836_{(0.002)}$	**$0.847_{(0.002)}$**
Tox21	$0.747_{(0.006)}$	**$0.799_{(0.005)}$**	$0.751_{(0.007)}$	**$0.800_{(0.006)}$**
HIV	$0.762_{(0.005)}$	**$0.821_{(0.005)}$**	$0.761_{(0.003)}$	**$0.823_{(0.002)}$**
MUV	$0.741_{(0.006)}$	**$0.834_{(0.009)}$**	$0.784_{(0.019)}$	**$0.834_{(0.010)}$**

请注意，加粗的数值表示表现最好的模型。所有指标均为平均受试者工作特征曲线下面积（AUC）。标准差以下标形式显示。

计算成本。虽然 B 样条 KAN 的参数数量也较少，但它们的局部控制点和插值步骤会引入额外开销。相比之下，傅里叶基提供平滑的全局近似，而没有这些负担。

Table 1 | Comparison of KA-GNNs with state-of-the-art GDL models across various molecular property prediction datasets

Model	BACE	BBBP	ClinTox	SIDER	Tox21	HIV	MUV
Number of molecules	1,513	2,039	1,478	1,427	7,831	41,127	93,808
Number of average atoms	65	46	50.58	65	36	46	43
Number of tasks	1	1	2	27	12	1	17
D-MPNN[46]	$0.809_{(0.006)}$	$0.710_{(0.003)}$	$0.906_{(0.007)}$	$0.570_{(0.007)}$	$0.759_{(0.007)}$	$0.771_{(0.005)}$	$0.786_{(0.014)}$
AttentiveFP[45]	$0.784_{(0.022)}$	$0.663_{(0.018)}$	$0.847_{(0.003)}$	$0.606_{(0.032)}$	$0.781_{(0.005)}$	$0.757_{(0.014)}$	$0.786_{(0.015)}$
N-GramRF[47]	$0.779_{(0.015)}$	$0.697_{(0.006)}$	$0.775_{(0.040)}$	$0.668_{(0.007)}$	$0.743_{(0.009)}$	$0.772_{(0.004)}$	$0.769_{(0.002)}$
N-GramXGB[47]	$0.791_{(0.013)}$	$0.691_{(0.008)}$	$0.875_{(0.027)}$	$0.655_{(0.007)}$	$0.758_{(0.009)}$	$0.787_{(0.004)}$	$0.748_{(0.002)}$
PretrainGNN[48]	$0.845_{(0.007)}$	$0.687_{(0.013)}$	$0.726_{(0.015)}$	$0.627_{(0.008)}$	$0.781_{(0.006)}$	$0.799_{(0.007)}$	$0.813_{(0.021)}$
GraphMVP[49]	$0.812_{(0.009)}$	$0.724_{(0.016)}$	$0.791_{(0.028)}$	$0.639_{(0.012)}$	$0.759_{(0.005)}$	$0.770_{(0.012)}$	$0.777_{(0.006)}$
MolCLR[50]	$0.824_{(0.009)}$	$0.722_{(0.021)}$	$0.912_{(0.035)}$	$0.589_{(0.014)}$	$0.750_{(0.002)}$	$0.781_{(0.005)}$	$0.796_{(0.019)}$
GEM[51]	$0.856_{(0.011)}$	$0.724_{(0.004)}$	$0.901_{(0.013)}$	$0.672_{(0.004)}$	$0.781_{(0.001)}$	$0.806_{(0.009)}$	$0.817_{(0.005)}$
Mol-GDL[28]	$0.863_{(0.019)}$	$0.728_{(0.019)}$	$0.966_{(0.002)}$	$0.831_{(0.002)}$	$0.794_{(0.005)}$	$0.808_{(0.007)}$	$0.675_{(0.014)}$
Uni-mol[52]	$0.857_{(0.002)}$	$0.729_{(0.006)}$	$0.919_{(0.018)}$	$0.659_{(0.013)}$	$0.796_{(0.005)}$	$0.808_{(0.003)}$	$0.821_{(0.013)}$
SMPT[53]	$0.873_{(0.015)}$	$0.734_{(0.003)}$	$0.927_{(0.002)}$	$0.676_{(0.050)}$	$0.797_{(0.001)}$	$0.812_{(0.001)}$	$0.822_{(0.008)}$
GNN-SKAN[40]	$0.747_{(0.009)}$	$0.676_{(0.014)}$	-	$0.614_{(0.005)}$	$0.747_{(0.005)}$	$0.786_{(0.015)}$	-
GraphKAN[39]	$0.823_{(0.011)}$	$0.731_{(0.017)}$	$0.984_{(0.003)}$	$0.837_{(0.001)}$	$0.753_{(0.007)}$	$0.711_{(0.016)}$	$0.715_{(0.014)}$
KA-GNNs[30]	$0.752_{(0.011)}$	$0.721_{(0.003)}$	$0.972_{(0.001)}$	$0.831_{(0.004)}$	$0.730_{(0.012)}$	$0.717_{(0.018)}$	$0.701_{(0.006)}$
KA-GCN	**$0.890_{(0.014)}$**	**$0.787_{(0.014)}$**	**$0.992_{(0.005)}$**	$0.842_{(0.001)}$	$0.799_{(0.005)}$	$0.821_{(0.005)}$	**$0.834_{(0.009)}$**
KA-GAT	$0.884_{(0.004)}$	$0.785_{(0.021)}$	$0.991_{(0.005)}$	**$0.847_{(0.002)}$**	$0.800_{(0.006)}$	**$0.823_{(0.002)}$**	$0.834_{(0.010)}$

Note that bold values indicate the best-performing models; underlined values indicate the second best. All metrics are average receiver operating characteristic-AUC and standard deviations are shown as subscripts.

Table 2 | Comparison of the performance of KA-GNN (KA-GCN and KA-GAT) models with three types of base function, including B-spline, polynomial and Fourier series

Dataset	KA-GCN models			KA-GAT models		
	B-spline	Polynomial	Fourier series	B-spline	Polynomial	Fourier series
BACE	$0.771_{(0.012)}$	$0.853_{(0.027)}$	**$0.890_{(0.014)}$**	$0.808_{(0.009)}$	$0.8319_{(0.007)}$	**$0.884_{(0.004)}$**
BBBP	$0.723_{(0.008)}$	$0.654_{(0.009)}$	**$0.787_{(0.014)}$**	$0.657_{(0.004)}$	$0.708_{(0.005)}$	**$0.785_{(0.021)}$**
ClinTox	$0.973_{(0.006)}$	$0.981_{(0.004)}$	**$0.992_{(0.005)}$**	$0.948_{(0.003)}$	$0.983_{(0.003)}$	**$0.991_{(0.005)}$**
SIDER	$0.824_{(0.003)}$	$0.832_{(0.006)}$	**$0.842_{(0.001)}$**	$0.825_{(0.004)}$	$0.836_{(0.001)}$	**$0.847_{(0.002)}$**
Tox21	$0.724_{(0.005)}$	$0.715_{(0.004)}$	**$0.799_{(0.005)}$**	$0.731_{(0.012)}$	$0.753_{(0.004)}$	**$0.800_{(0.006)}$**
HIV	$0.753_{(0.007)}$	$0.804_{(0.010)}$	**$0.821_{(0.005)}$**	$0.744_{(0.018)}$	$0.818_{(0.006)}$	**$0.823_{(0.002)}$**
MUV	$0.638_{(0.008)}$	$0.787_{(0.012)}$	**$0.834_{(0.009)}$**	$0.792_{(0.013)}$	$0.797_{(0.011)}$	**$0.834_{(0.010)}$**

Note that bold values indicate the best-performing models. All metrics are average receiver operating characteristic-AUC. Standard deviations are shown as subscripts.

highlighting Fourier KANs' enhanced ability to capture chemical structure representations.

To investigate the architectural drivers behind KA-GNNs' performance, we conducted an extensive ablation study examining input normalization, KAN modularity, graph construction (for example, non-covalent bonds), regression capabilities, scalability and hyperparameter sensitivity (Supplementary Tables 3–8).

Notably, we tested six normalization schemes for node and edge features. KA-GNNs maintained strong performance across all, underscoring their robustness to input scale variation.

Efficiency of Fourier-based KAN in GNNs

As shown in Fig. 2a,b, Fourier-based KAN models consistently achieve lower runtime than both B-spline KANs and MLP-based GCN/GATs. This efficiency arises from the global nature of sinusoidal basis functions, which enable compact function approximation with fewer trainable parameters.

Unlike MLPs with large dense weight matrices, KANs use fixed basis functions with lightweight learnable coefficients, notably reducing

Table 3 | Comparison of GCN/GAT models and KA-GCN/KA-GAT with Fourier series

Dataset	GCN	KA-GCN	GAT	KA-GAT
BACE	$0.835_{(0.014)}$	**$0.890_{(0.014)}$**	$0.834_{(0.012)}$	**$0.884_{(0.004)}$**
BBBP	$0.735_{(0.011)}$	**$0.787_{(0.014)}$**	$0.707_{(0.007)}$	**$0.785_{(0.021)}$**
ClinTox	$0.979_{(0.004)}$	**$0.992_{(0.005)}$**	$0.983_{(0.006)}$	**$0.991_{(0.005)}$**
SIDER	$0.834_{(0.001)}$	**$0.842_{(0.001)}$**	$0.836_{(0.002)}$	**$0.847_{(0.002)}$**
Tox21	$0.747_{(0.006)}$	**$0.799_{(0.005)}$**	$0.751_{(0.007)}$	**$0.800_{(0.006)}$**
HIV	$0.762_{(0.005)}$	**$0.821_{(0.005)}$**	$0.761_{(0.003)}$	**$0.823_{(0.002)}$**
MUV	$0.741_{(0.006)}$	**$0.834_{(0.009)}$**	$0.784_{(0.019)}$	**$0.834_{(0.010)}$**

Note that bold values indicate the best-performing models. All metrics are average receiver operating characteristic-AUC. Standard deviations are shown as subscripts.

computational cost. Although B-spline KANs also have lower parameter counts, their localized control points and interpolation steps introduce overhead. By contrast, Fourier bases offer smooth, global approximations without such burdens.

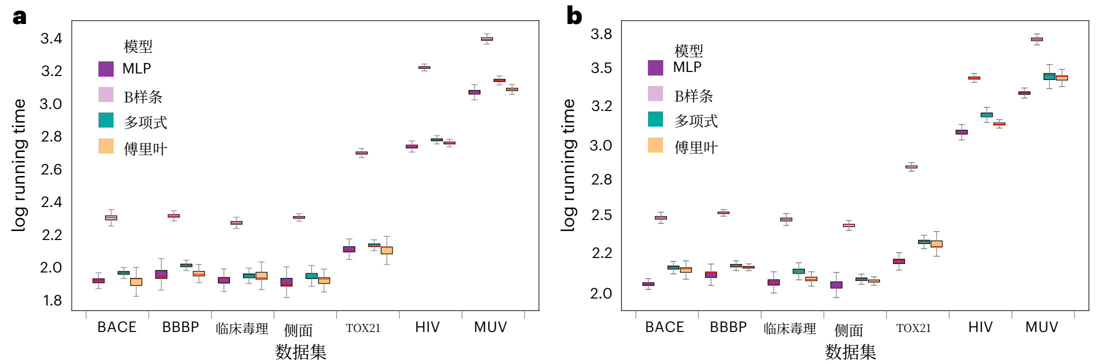

图 2 | 不同基函数下 MLP 基模型和 KA-GNN 在七个基准数据集上的效率比较。a、KA-GCN 模型。b、KA-GAT 模型。箱线图显示了使用不同基函数的 MLP 基模型和 KA-GNN 的运行时间（以秒为单位）的对数分布跨七个数据集。每个箱子代表四分位距，中心线表示中位数，胡须延伸至1.5× 四分位距。单个数据点（n = 3）以点的形式叠加。

基于傅里叶的KA-GNN在效率上也优于传统的GCN和GAT。虽然GCN的扩展性取决于节点和层的数量，GAT则取决于注意力头的数量，傅里叶-KAN通过消除冗余参数矩阵简化了消息传递。这加快了收敛速度并降低了训练时间，使其非常适合大规模图学习。

KA-GAT分析的解释

KAN模块的可解释性是一个关键优势。为了展示这一点，我们分析了KA-GAT对分子Fc1ccc(NC(=O)c2ncc(cc2)C#N)cc1[C@]1(N=C(OCC1)N)C的预测。图3a和3b显示了其二维和三维结构（通过MolView）作为化学参考。图3c提供了一个显著性图，突出显示了原子和键级别的贡献。将分子建模为图G= (V, E)，其特征矩阵为f_V和f_E，并使用一个针对类别c的训练模型f_θ，输出分数为$s= f^\theta(G, f^V, f^E)_c$。利用输入梯度，$L^1$范数量化每个输入的影响：

$$\text{Saliency}(v) = \sum_{k=1}^{d_1}\left|\frac{\partial s}{\partial \mathbf{f}_v[k]}\right|, \quad \text{Saliency}(e) = \sum_{k=1}^{d_2}\left|\frac{\partial s}{\partial \mathbf{f}_e[k]}\right|.$$

这里，d_1和d_2分别是节点特征和边特征的维度，$f_v[k]$ 和$f_e[k]$ 是f_v 和f_e的第k个元素。这些梯度衡量预测对输入扰动的敏感性。

为了提取关键子结构，我们保留归一化显著性超过第40百分位的节点和边，从而得到一个包含16个节点、11条边的高归因子子图。功能基团使用SMARTS模式定义，基团级显著性计算如下：

$$\text{Saliency}_{\text{group}}(g) = \frac{1}{|V_g|}\sum_{u\in V_g}\text{Saliency}_V(u),$$

其中 $V_g \subseteq V$ 是 g 组中的原子。在这种情况下，氟（–F）得分最高（2.9535），其次是酰胺（–CONH–，2.4133）、芳香环（2.3251）和胺（–NH–，1.1992），反映了模型对吸电子和共振稳定基团的关注。

图 3d 显示了来自 KA-GAT 调整的 GNNExplainer 的子图重要性。软掩码 $m_V \in [0,1]^{|V|}$、$m_E \in [0,1]^{|E|}$ 被逐元素应用于特征。损失在输出保真度和稀疏性之间进行平衡：

$$\mathcal{L} = \left\| f_\theta(G, f_V, f_E) - f_\theta(G, f_V \odot m_V, f_E \odot m_E) \right\|^2 + \lambda_V \parallel m_V \parallel_1 + \lambda_E \parallel m_E \parallel_1,$$

其中 λ_V 和 λ_E 是正则化项。由此得到的子图 $G' \subseteq G$ 捕捉了最具影响力的原子和键。

总之，KA-GAT能够识别化学上相关的特征，如氟化基团和酰胺基团，从而产生可解释且与领域一致的预测，支持其在药物发现中的使用。

讨论

我们的KA-GNNs已经将KAN模块协同整合到GNN架构中，并具有三个核心创新。首先，在分子图构建中，我们通过基于截断距离引入边来结合共价和非共价相互作用。这种方法使KA-GNN能够捕捉超出传统共价键的更广泛的分子相互作用，从而对分子结构及其性质有更全面的理解。其次，我们将KAN集成到GNN框架中，显著减少参数数量，同时增强可解释性和表达能力。与传统的基于MLP的信息传递方案不同，KAN提供了更灵活的函数逼近机制，从而在更少可训练参数的情况下提高预测准确性。第三，我们在KAN中引入了基于傅里叶的激活函数，取代原始的B样条函数。傅里叶级数被广泛用于信号处理和函数逼近，使得

即使性能表现有希望，我们的 KA-GNN 仍然可以进一步改进，特别是学习到的基于傅里叶的表示与显式化学和生物学原理的对齐仍然是一个巨大的挑战。虽然显著性图和子图重要性分析已经表明我们的 KA-GNN 能够有效识别关键分子子结构、关键原子相互作用以及功能相关的子图，但缺乏将单个或与频率相关的傅里叶函数与物理或化学意义联系起来的深入特征重要性分析。事实上，使用基于 KAN 的剪枝方法对学习到的傅里叶函数进行特征重要性分析时，我们没有获得任何“有意义”的结果。未来研究的一个令人兴奋的方向是引入领域特定约束、利用对比学习策略，或应用物理化学知情的傅里叶函数，以提高 KA-GNN 学习表示的可解释性。

适当设计的傅里叶分量可能具有巨大的潜力，有效地表征和表示分子的深层物理和化学特性。从物理和化学的角度来看，量子力学模型，如薛定谔方程、密度泛函理论、分子轨道理论等，都涉及

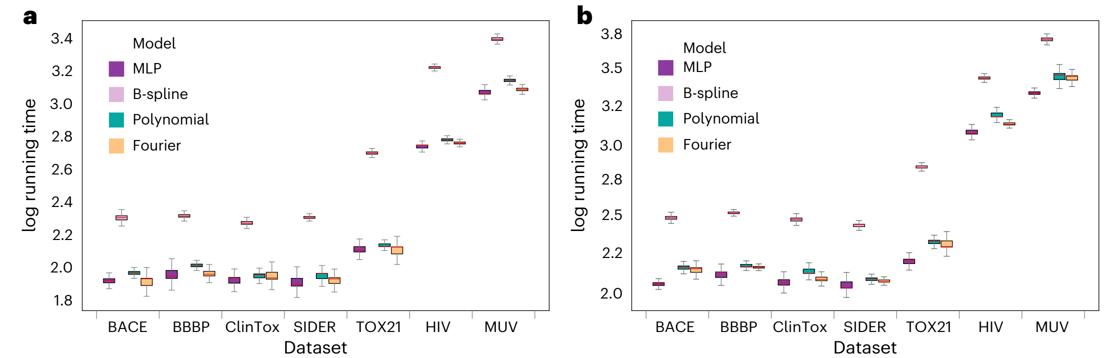

Fig. 2 | Efficiency comparison of MLP-based models and KA-GNNs with different basis functions across seven benchmark datasets. a, KA-GCN models. **b**, KA-GAT models. Boxplots showing the distribution of $\log_{10}$ running times (in seconds) for MLP-based models and KA-GNNs using different basis functions across seven datasets. Each box represents the interquartile range, the centre line indicates the median and whiskers extend to 1.5× interquartile range. Individual data points (n = 3) are overlaid as dots.

Fourier-based KA-GNNs also outperform traditional GCNs and GATs in efficiency. While GCNs scale with node and layer count, and GATs with attention heads, Fourier-KAN simplifies message passing by eliminating redundant parameter matrices. This accelerates convergence and lowers training time, making it highly suitable for large-scale graph learning.

Interpretation of KA-GAT analysis

The interpretability of the KAN module is a key advantage. To demonstrate this, we analyse KA-GAT's prediction for the molecule Fc1ccc(NC(=O)c2ncc(cc2)C#N)cc1[C@]1(N=C(OCC1)N)C. Figure 3a,b shows its 2D and 3D structures (via MolView) as chemical references. Figure 3c provides a saliency map highlighting atom- and bond-level contributions. Modelling the molecule as a graph $G = (V, E)$ with feature matrices f_V and f_E, and using a trained model f_θ targeting class c, the output score is $s = f_\theta(G, f_V, f_E)_c$. Using input gradients, the L^1 norm quantifies each input's influence:

$$\text{Saliency}(v) = \sum_{k=1}^{d_1}\left|\frac{\partial s}{\partial \mathbf{f}_v[k]}\right|, \quad \text{Saliency}(e) = \sum_{k=1}^{d_2}\left|\frac{\partial s}{\partial \mathbf{f}_e[k]}\right|.$$

Here, d_1 and d_2 is the dimensional of node feature and edge feature, respectively, and $\mathbf{f}_v[k]$ and $\mathbf{f}_e[k]$ is the kth element of $\mathbf{f}_v$ and $\mathbf{f}_e$. These gradients measure prediction sensitivity to input perturbations.

To extract key substructures, we retain nodes and edges above the 40th percentile of normalized saliency, yielding a 16-node, 11-edge high-attribution subgraph. Functional groups are defined using SMARTS patterns, with group-level saliency computed as:

$$\text{Saliency}_{\text{group}}(g) = \frac{1}{|V_g|}\sum_{u\in V_g}\text{Saliency}_V(u),$$

where $V_g \subseteq V$ are the atoms in group g. In this case, fluoro (–F) ranks highest (2.9535), followed by amide (–CONH–, 2.4133), aromatic rings (2.3251) and amine (–NH–, 1.1992), reflecting the model's attention to electron-withdrawing and resonance-stabilizing groups.

Figure 3d shows subgraph importance from a KA-GAT-adapted GNNExplainer. Soft masks $m_V \in [0, 1]^{|V|}$, $m_E \in [0, 1]^{|E|}$ are applied element-wise to features. The loss balances output fidelity and sparsity:

$$\mathcal{L} = \left\| f_\theta(G, f_V, f_E) - f_\theta(G, f_V \odot m_V, f_E \odot m_E) \right\|^2 + \lambda_V \parallel m_V \parallel_1 + \lambda_E \parallel m_E \parallel_1$$

where λ_V and λ_E are regularization terms. The resulting subgraph $G' \subseteq G$ captures the most influential atoms and bonds.

In summary, KA-GAT identifies chemically relevant features, such as fluoro and amide groups, yielding interpretable and domain-aligned predictions that support its use in drug discovery.

Discussion

Our KA-GNNs have synergistically incorporated KAN module into GNN architecture and have three core innovations. First, in molecular graph construction, we incorporate both covalent and non-covalent interactions by introducing edges based on a cut-off distance. This approach allows KA-GNN to capture a broader range of molecular interactions beyond traditional covalent bonds, leading to a more comprehensive understanding of molecular structures and their properties. Second, we integrate the KAN into the GNN framework, notably reducing the number of parameters while enhancing interpretability and expressive power. Unlike traditional MLP-based message-passing schemes, KAN offers a more flexible function approximation mechanism, leading to improved predictive accuracy with fewer trainable parameters. Third, we introduce a Fourier-based activation function in KAN, replacing the original B-spline functions. Fourier series, widely used in signal processing and function approximation, enable KA-GNN to better capture high-frequency patterns in graph-structured data. By learning and optimizing Fourier coefficients, KA-GNN achieves higher accuracy and stability, particularly in modelling complex molecular interactions and biochemical systems.

Even with the promising performance, our KA-GNNs can be further improved, in particular, the alignment of the learned Fourier-based representations with explicit chemical and biological principles remains a great challenge. Although saliency maps and subgraph importance analysis have demonstrated that our KA-GNN effectively identifies key molecular substructures, critical atomic interactions and functionally relevant subgraphs, an insightful feature importance analysis that connects an individual or frequency-related Fourier function to physical or chemical meanings is absent. In fact, we do not obtain any 'meaningful' results using KAN-based pruning approach for feature importance analysis of our learned Fourier functions. An exciting direction for future research involves incorporating domain-specific constraints, leveraging contrastive learning strategies, or applying physico-chemically informed Fourier functions to improve the interpretability of KA-GNN's learned representations and their alignment with expert domain knowledge.

The properly designed Fourier components may have huge potential to effectively characterize and represent deep physical and chemical properties of molecules. From the physical and chemical perspective, quantum mechanics models, such as Schrodinger equation, density function theory, molecular orbital theory and so on, all involve

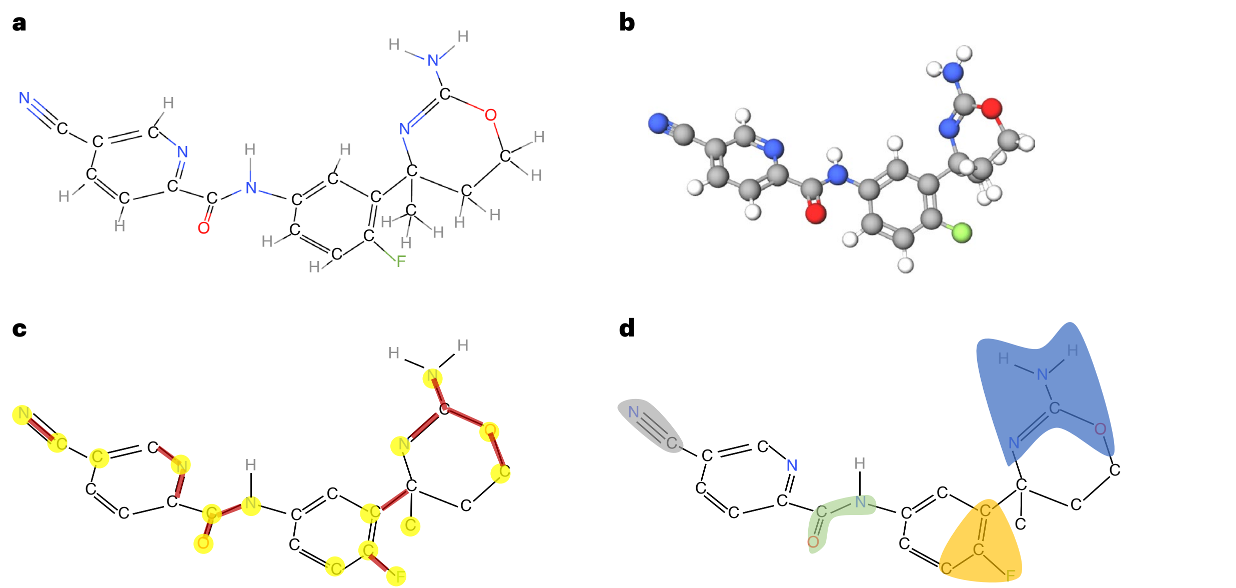

图 3 | KA-GAT 对分子 Fc1ccc(NC(=O)c2ncc(cc2) C#N)cc1[C@]1(N=C(OCC1)N)C 的解释。a，使用 MolView 可视化的二维结构。b，通过展开手性中心 [C@] 并优化分子几何生成的三维结构。c，突出显示的重要性图 对模型预测影响最大的原子和化学键（颜色越亮表示重要性越高）。d，通过基于掩码的GNNExplainer得出的子图重要性，突出影响预测的关键功能基团。

谐波或波动性质在数学上表示为一系列傅里叶型函数。从学习的角度来看，球谐函数已经被广泛应用于图深度学习（GDL）模型中，用于描述分子的等变性质。这些球谐基函数恰好是傅里叶型函数。最后，傅里叶分量不仅可以用于频率扰动分析，还可以用于表征各种量子描述符，例如最高占据分子轨道（HOMO）与最低未占分子轨道（LUMO）能隙、原子部分电荷或对称适应基函数。这些研究可能有助于弥合图表示与分子功能和性质之间的差距。

结论

在本研究中，我们提出了非平凡的基于KAN的GNN。我们的KA-GNN利用KAN的独特能力在三个核心组件上优化GNN架构，包括节点嵌入、信息传递和读出。我们开发了基于傅里叶级数的KAN模型，并提供了其逼近能力的严格理论证明。我们进一步比较了不同的预激活单变量函数在模型性能和计算效率方面的表现，并确定傅里叶级数是最有效的选择。在广泛使用的分子属性预测基准上的大量实验表明，我们的KA-GNN始终优于传统的GNN模型。这项工作不仅凸显了KA-GNN在分子任务中的潜力，也为通用非欧几里得数据分析提供了一个GDL框架。

方法

Kolmogorov–Arnold 网络及其理论背景

Kolmogorov–Arnold 表示定理（或叠加定理）是实分析和逼近理论领域的一个里程碑[54]。它指出，每一个多变量连续函数都可以表示为一元连续函数相加的叠加形式。这个定理不仅解决了希尔伯特的第十三问题本身，而且将其推广到了更广泛的形式。

弗拉基米尔·阿诺尔德和安德烈·柯尔莫哥洛夫的研究证明，任意多元连续函数 f 可以表示为有限的单变量连续函数的组合以及加法的二元运算。更具体地说，

$$f(x_1,\dots,x_n)=\sum_{q=0}^{2n+1}\Phi_q\left(\sum_{p=1}^{n}\phi_{q,p}(x_p)\right), \qquad (2)$$

其中 n 表示多元函数 f 的变量个数，$\Phi_q:\mathbb{R}\to\mathbb{R}$ 和 $\phi_{q,p}:[0,1]\to\mathbb{R}$ 是连续的单变量函数。

KAN。受Kolmogorov–Arnold表示定理的启发，一项研究提出了一种名为KAN的新型深度学习架构，作为MLP的有前途的替代方案。为了增强表示能力并利用现代技术（例如反向传播）来训练网络，KAN对Kolmogorov–Arnold表示定理中的设置进行了推广：

- KAN 并不遵循原始的深度-2 宽度（2n +1）表示；相反，它推广到任意宽度和深度。具体来说，设第 l 层的激活值表示为 $\mathbf{x}^{(l)} := (x_1^{(l)}, x_2^{(l)}, \cdots, x_{n_l}^{(l)})$，其中 n_l 为第 l 层的宽度。第 l 层的激活值 +1 则简单地为所有传入后激活值的和：

$$x_j^{(l+1)}=\sum_{i=1}^{n_l}\phi_{j,i}^{(l)}\left(x_i^{(l)}\right), \quad j=1,\dots,n_{l+1}. \qquad (3)$$

这里，$\phi^{(l)}_{j,i}$ 对于 i =1, …, n_l 和 j =1, …, n_{l+1} 是层 l 中的前激活函数。这些函数在 KAN 中的作用类似于方程 (2) 中内函数 $\phi_{q,p}$ 的作用。• 虽然对 Kolmogorov–Arnold 表示定理的许多构造性证明表明方程 (2) 中的内函数 $\phi_{q,p}$ 是高度不光滑的，[55,56], KAN 选择平滑函数作为前激活函数 $\phi^{(l)}_{j,i}$ 以便于反向传播。在参考文献 29 中，作者根据 B 样条选择函数。

傅里叶级数 KAN 模型。为了优化网络并避免复杂的计算，我们建议如公式（3）中所示，将傅里叶级数用作 KAN[37] 的预激活函数：

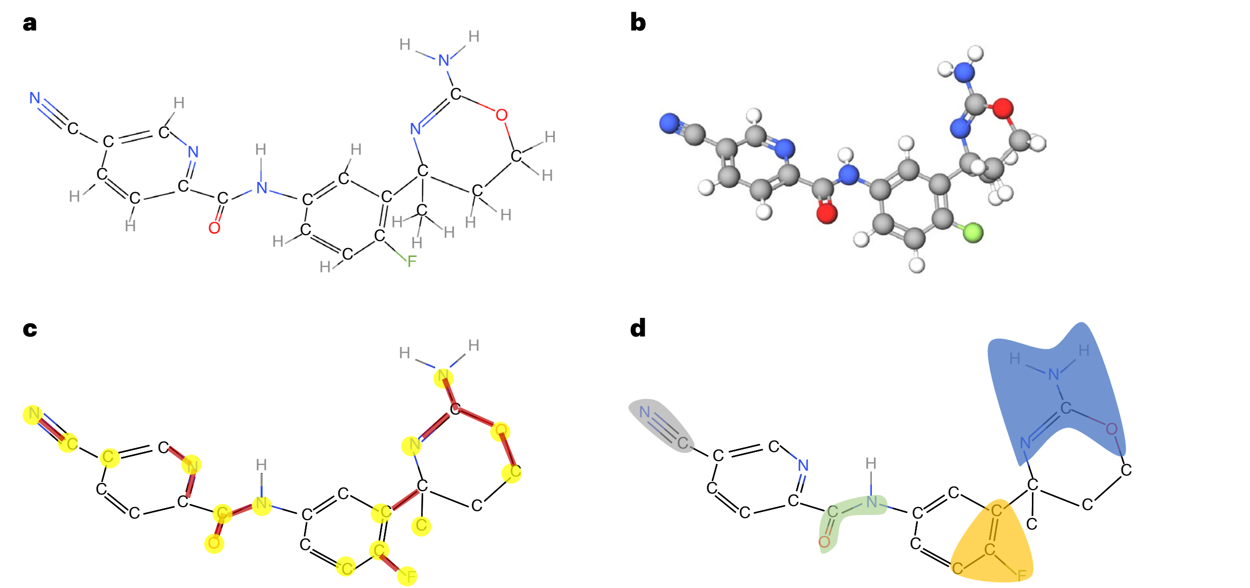

Fig. 3 | KA-GAT interpretation of the molecule Fc1ccc(NC(=O)c2ncc(cc2) C#N)cc1[C@]1(N=C(OCC1)N)C. a, Two-dimensional structure visualized using MolView. **b**, Three-dimensional structure generated by expanding the chiral centre [C@] and optimizing molecular geometry. **c**, Saliency map highlighting the atoms and bonds most influential to the model's prediction (a brighter colour indicates higher importance). **d**, Subgraph importance derived from a mask-based GNNExplainer, highlighting key functional groups influencing the prediction.

harmonic or wave-like properties that are mathematically represented as a series of Fourier-type functions. From the learning perspective, spherical harmonics have already been widely used in GDL models to characterize equivariant properties of molecules. These spherical harmonics basis functions are exactly Fourier-type functions. Finally, the Fourier components can be used in not only frequency perturbation analysis, but also characterizing various quantum descriptors such as highest occupied molecular orbital and lowest unoccupied molecular orbital gaps, atomic partial charges or symmetry-adapted basis functions. These studies may help to bridge the gap between graph representations and molecular functions and properties.

Conclusion

In this study, we propose the non-trivial KAN-based GNNs. Our KA-GNNs use KAN's unique power to optimize GNN architectures at three core components, including node embedding, message passing and readout. We develop a Fourier-series-based KAN model and provide a rigorous theoretical justification of its approximation capability. We further compare different choices of pre-activation univariate functions in terms of both model performance and computational efficiency, and identify the Fourier series as the most effective choice. Extensive experiments on widely used molecular property prediction benchmarks demonstrate that our KA-GNNs consistently outperform traditional GNN models. This work not only highlights the potential of KA-GNNs for molecular tasks, but also contributes a GDL framework for general non-Euclidean data analysis.

Methods

Kolmogorov–Arnold networks and their theoretical background

The Kolmogorov–Arnold representation theorem (or superposition theorem) is a milestone in the field of real analysis and approximation theory[54]. It states that every multivariate continuous function can be represented as a superposition of the addition of continuous functions of one variable. This theorem not only solves Hilbert's 13th problem itself but also generalizes it to a broader form.

Vladimir Arnold and Andrey Kolmogorov's studies[54] prove that arbitrary multivariate continuous function f can be written as a finite composition of continuous functions of a single variable and the binary operation of addition. More specifically,

$$f(x_1,\dots,x_n)=\sum_{q=0}^{2n+1}\Phi_q\left(\sum_{p=1}^{n}\phi_{q,p}(x_p)\right), \qquad (2)$$

where n denotes the number of variables of the multivariate function f, and $\Phi_q:\mathbb{R}\to\mathbb{R}$ and $\phi_{q,p}:[0,1]\to\mathbb{R}$ are continuous single-variable functions.

KAN. Inspired by the Kolmogorov–Arnold representation theorem, one study[29] proposed a new deep learning architecture called the KAN as a promising alternative to the MLP. To enhance the representational capacity and leverage modern techniques (for example, backpropagation) for training the networks, KAN generalizes the setting in Kolmogorov–Arnold representation theorem:

- KAN does not adhere to the original depth-2 width $(2n+1)$ representation; instead, it generalizes to arbitrary widths and depths. Specifically, let the activation values in layer l be denoted by $\mathbf{x}^{(l)} := (x_1^{(l)}, x_2^{(l)}, \cdots, x_{n_l}^{(l)})$, where n_l is the width of layer l. The activation value in layer $l+1$ is then simply the sum all incoming postactivations:

$$x_j^{(l+1)}=\sum_{i=1}^{n_l}\phi_{j,i}^{(l)}\left(x_i^{(l)}\right), \quad j=1,\dots,n_{l+1}. \qquad (3)$$

  Here, $\phi_{j,i}^{(l)}$ for $i=1,\dots,n_l$ and $j=1,\dots,n_{l+1}$ are the pre-activation functions in layer l. The roles of these functions in KAN are analogous to the roles of the inner functions $\phi_{q,p}$ in equation (2).
- Although many constructive proofs of the Kolmogorov–Arnold representation theorem indicate that the inner functions $\phi_{q,p}$ in equation (2) are highly non-smooth[55,56], KAN opts for smooth functions as pre-activation functions $\phi_{j,i}^{(l)}$ to facilitate backpropagation. In ref. 29 the authors selected functions on the basis of B-splines.

Fourier-series KAN model. To optimize the network and avoid complex calculations, we propose to use Fourier series as the pre-activation functions for KAN[37] as in equation (3):

$$\phi_{j,i}^{(l)}(x)=\sum_{k=1}^{K}\left(A_{k,j,i}^{(l)}\cos(kx)+B_{k,j,i}^{(l)}\sin(kx)\right),\qquad(4)$$

其中 K 是谐波的数量，A(l) k, j,i 和 B(l) k, j,i 是可学习的参数，最初从正态分布 $\mathcal{N}\left(0, \frac{1}{n_{l+1}\times K}\right)$ 中采样。

因此，层 l 中第 j 个神经元的激活值 +1 可以通过以下方式获得：

$$x_j^{(l+1)}=\sum_{i=1}^{n_l}\sum_{k=1}^{K}\left(A_{k,j,i}^{(l)}\cos\left(kx_i^{(l)}\right)+B_{k,j,i}^{(l)}\sin\left(kx_i^{(l)}\right)\right),\qquad(5)$$

其中$x^{(l)} = (x_1^{(l)}, x_2^{(l)}, \cdots, x_{n_l}^{(l)})$ 表示KAN第l层的激活值输入向量，$x^{(l+1)} = (x_1^{(l+1)}, x_2^{(l+1)}, \cdots, x_{n_{l+1}}^{(l+1)})$ 表示输出向量。方程（5）可以简洁地表示为：$x^{(l+1)} = \mathrm{KAN}_l(x^{(l)})$，其中$\mathrm{KAN}_l(\cdot)$表示上述第l层的KAN函数。该网络被集成到GNNs中，形成了一种名为KA-GNNs的架构。

基于KAN的GNN模型

在我们的 KA-GNN 模型中，一个分子表示为图 G = (V, E)，其中 V 表示节点集合，E 表示边集合。每个节点 v ∈ V 关联一个特征向量 $f_v \in \mathbb{R}^{d_1}$，每条边 uv ∈ E 关联一个特征向量 $f_{uv} \in \mathbb{R}^{d_2}$。这里，uv 表示连接节点 u 和 v 的边，$d_1$ 和 d_2 分别表示节点特征和边特征的维度。每个节点代表一个原子，如果任意两个原子的距离在截止距离内，则形成一条边（在我们的模型中，截止距离设为 5 Å）。图 1a 展示了分子内部的复杂相互作用，突出显示了共价键（实线）和非共价截止键（虚线），在我们的 KA-GNN 模型中，距离小于 5 Å 的非共价键也被考虑在内。我们的原子特征，包括原子序数、半径和电负性，是利用 Rdkit 提取的，并按照晶体图卷积神经网络[43]和路径复杂神经网络的方法获得。

KA-GCN 模型。KA-GCN 通过在模型的关键阶段引入 KAN 层，取代传统的基于 MLP 的转换，从而扩展了标准 GCN。与采用固定激活函数（例如 ReLU）的传统 MLP 不同，KAN 动态学习非线性转换，从而实现更具表现力的特征传播，并提高对复杂分子图的适应能力。

(1) 节点嵌入初始化

KA-GCN没有使用简单的线性变换，而是通过KAN结合节点和邻居边特征来增强初始节点嵌入：

$$h_v^{(0)} := \mathrm{KAN}\left(\mathbf{f}_v \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)}\mathbf{f}_{vu}\right)\right),\qquad(6)$$

其中 N(v) 表示相邻节点的集合（不包括节点 v 本身），|N(v)| 表示相邻节点的总数，⊕ 表示向量连接。

(2) KAN 增强型消息传递

在每一层 l，KA-GCN 通过使用残差 KAN 变换从邻居聚合信息来优化节点表示：

$$h_v^{(l+1)} := h_v^{(l)} + \mathrm{KAN}\left(h_v^{(l)} \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)} h_u^{(l)}\right)\right).\qquad(7)$$

在这里，KAN 替代了标准 GCN 中的线性变换，动态调整特征交互，而不是应用固定权重的求和。

(3) 输出和预测

在经过 L 层消息传递层之后，我们使用 KAN 进行图级表示聚合：

$$\hat{y} := \mathrm{KAN}\left(\frac{1}{|V|}\sum_{v\in V} h_v^{(L)}\right).$$

与简单的 MLP 分类器相比，基于 KAN 的读出动态地学习分子图表示中的复杂模式。

对于分类任务，我们使用交叉熵损失：

$$\mathcal{L} := -\sum_i \left(y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right).$$

总体而言，KA-GCN通过允许自适应的数据驱动特征变换来增强表达能力，从而解决了标准GCN中固定激活函数的僵化问题。

KA-GAT 模型。类似于 KA-GCN，KA-GAT 模型将 GAT 中基于 MLP 的变换替换为 KAN，从而实现自适应注意机制。标准 GAT 使用依赖于静态参数化函数的固定注意机制，而 KA-GAT 能够学习更灵活、依赖数据的注意机制。

(1) 自适应初始化

节点和边的初始嵌入通过 KAN 层进行增强：

$$\begin{aligned} h_v^{(0)} &:= \mathrm{KAN}\left(\mathbf{f}_v \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)}\mathbf{f}_{vu}\right)\right),\\ h_{vu}^{(0)} &:= \mathrm{KAN}\left(W_{hv}\mathbf{f}_v + W_e\mathbf{f}_{vu} + W_{tv}\mathbf{f}_u\right). \end{aligned}\qquad(8)$$

KAN 实现了原子级和键级特征的非线性融合，从而允许更具表现力的初始表示。这里 W_{hv}、W_e 和 W_{tv} 是用于将输入特征缩放到相同维度的权重矩阵。

(2) KAN增强注意力机制

KA-GAT 不是进行固定的注意力计算，而是通过以下方式学习自适应注意力分数：

$$\begin{aligned} z_v^{(l+1)} &:= Wh_v^{(l)}, \quad z_u^{(l+1)} := Wh_u^{(l)},\\ \alpha_{vu} &:= \mathrm{Softmax}\left(h_{vu}^{(l)}\right), \quad m_{vu}^{(l+1)} := z_u^{(l+1)}\cdot\alpha_{vu},\\ m_v^{(l+1)} &:= z_v^{(l+1)} + \frac{1}{|N(v)|}\sum_{u\in N(v)} m_{vu}^{(l+1)}. \end{aligned}\qquad(9)$$

这里 W 是一个数值矩阵。在此使用 KAN 可以实现自适应、依赖上下文的注意力分数，这与使用固定权重注意力机制的标准 GAT 不同。

(3) 基于KAN的特征更新

在计算基于注意力的信息后，我们应用KAN进行特征优化：

$$h_v^{(l+1)} := \mathrm{KAN}\left(m_v^{(l+1)}\right), \quad h_{vu}^{(l+1)} := \mathrm{KAN}\left(h_{vu}^{(l)} + h_u^{(l)} + h_v^{(l)}\right).\qquad(10)$$

这确保了节点和边的特征以非线性、数据驱动的方式进行更新。

(4) 读出和最终预测

类似于 KA-GCN，KA-GAT 使用基于 KAN 的图级特征聚合，允许对分子图进行灵活表示。

$$\phi_{j,i}^{(l)}(x)=\sum_{k=1}^{K}\left(A_{k,j,i}^{(l)}\cos(kx)+B_{k,j,i}^{(l)}\sin(kx)\right),\qquad(4)$$

where K is the number of harmonics, and $A_{k,j,i}^{(l)}$ and $B_{k,j,i}^{(l)}$ are learnable parameters initially sampled from a normal distribution $\mathcal{N}\left(0, \frac{1}{n_{l+1}\times K}\right)$.

Consequently, the activation value at the jth neuron in layer $l+1$ can be obtained by,

$$x_j^{(l+1)}=\sum_{i=1}^{n_l}\sum_{k=1}^{K}\left(A_{k,j,i}^{(l)}\cos\left(kx_i^{(l)}\right)+B_{k,j,i}^{(l)}\sin\left(kx_i^{(l)}\right)\right),\qquad(5)$$

where $\mathbf{x}^{(l)} = (x_1^{(l)}, x_2^{(l)}, \dots, x_{n_l}^{(l)})$ denotes the input vector of activation values in layer l of the KAN, and $\mathbf{x}^{(l+1)} = (x_1^{(l+1)}, x_2^{(l+1)}, \dots, x_{n_{l+1}}^{(l+1)})$ represents the output vector. Equation (5) can be concisely expressed as: $\mathbf{x}^{(l+1)} = \mathrm{KAN}_l(\mathbf{x}^{(l)})$, where $\mathrm{KAN}_l(\cdot)$ denotes the above KAN function in layer l. This network is integrated into GNNs, resulting in a architecture named KA-GNNs.

KAN-based GNN models

In our KA-GNN models, a molecule is represented as a graph $G = (V, E)$, where V denotes the set of nodes and E denotes the set of edges. Each node $v \in V$ is associated with a feature vector $\mathbf{f}_v \in \mathbb{R}^{d_1}$, and each edge $uv \in E$ is associated with a feature vector $\mathbf{f}_{uv} \in \mathbb{R}^{d_2}$. Here, uv denotes the edge that connecting node u and v, and d_1 and d_2 denote the dimensions of node feature and edge feature, respectively. Each node represents an atom and an edge is formed among any two atoms if their distance is within a cut-off distance (in our model, we use cut-off distance as 5 Å). Figure 1a illustrates the complex interactions within the molecule, highlighting both the covalent bonds (solid lines) and non-covalent cut-off bonds (dashed lines) with distances less than 5 Å are considered in our KA-GNN model. Our atomic features, comprising atomic number, radius and electronegativity, are derived using Rdkit, following the approach in crystal graph convolutional neural networks[43] and path complex neural networks[57]. Each node $v \in V$ is associated with a feature vector $\mathbf{f}_v$, which is a 92-dimensional vector composed of one-hot encoded representations of atomic properties, following the approach described in ref. 43. Similarly, each edge $uv \in E$ is associated with a feature vector $\mathbf{f}_{uv}$, which is a 21-dimensional vector incorporating both chemical and geometrical information of the bond uv. The edges in our model can be classified into two types, that is, covalent bonds and non-covalent bonds, with different initial features. Detailed descriptions of the feature vectors are provided in the Supplementary Information.

KA-GCN model. KA-GCN extends the standard GCN by incorporating KAN layers at key stages of the model, replacing traditional MLP-based transformations. Unlike conventional MLPs, which apply fixed activation functions (for example, ReLU), KAN dynamically learns nonlinear transformations, allowing for more expressive feature propagation and improved adaptability to complex molecular graphs.

(1) Node embedding initialization

Instead of using a simple linear transformation, KA-GCN enhances initial node embeddings by incorporating both node and neighbourhood edge features through KAN:

$$h_v^{(0)} := \mathrm{KAN}\left(\mathbf{f}_v \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)}\mathbf{f}_{vu}\right)\right),\qquad(6)$$

where $N(v)$ represents the set of neighbouring nodes (excluding v itself) with $|N(v)|$ the total number of the neighbouring nodes and ⊕ represents vector concatenation.

(2) KAN-enhanced message passing

At each layer l, KA-GCN refines node representations by aggregating information from neighbours using a residual KAN transformation:

$$h_v^{(l+1)} := h_v^{(l)} + \mathrm{KAN}\left(h_v^{(l)} \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)} h_u^{(l)}\right)\right).\qquad(7)$$

Here, KAN replaces the linear transformation in standard GCNs, dynamically adjusting feature interactions instead of applying a fixed-weight sum.

(3) Readout and prediction

After L message-passing layers, we use KAN for graph-level representation aggregation:

$$\hat{y} := \mathrm{KAN}\left(\frac{1}{|V|}\sum_{v\in V} h_v^{(L)}\right).$$

Compared with a simple MLP classifier, the KAN-based readout dynamically learns complex patterns in the molecular graph representation.

For classification tasks, we use the cross-entropy loss:

$$\mathcal{L} := -\sum_i \left(y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right).$$

Overall, KA-GCN enhances expressiveness by allowing adaptive, data-driven feature transformations, addressing the rigidity of fixed activation functions in standard GCNs.

KA-GAT model. Similar to KA-GCN, the KA-GAT model replaces MLP-based transformations in GAT with KAN, enabling adaptive attention mechanisms. Standard GAT uses a fixed attention mechanism that relies on static parameterized functions, whereas KA-GAT learns more flexible, data-dependent attention mechanisms.

(1) Adaptive initialization

The initial embeddings for nodes and edges are enhanced using KAN layers:

$$\begin{aligned} h_v^{(0)} &:= \mathrm{KAN}\left(\mathbf{f}_v \oplus \left(\frac{1}{|N(v)|}\sum_{u\in N(v)}\mathbf{f}_{vu}\right)\right),\\ h_{vu}^{(0)} &:= \mathrm{KAN}\left(W_{hv}\mathbf{f}_v + W_e\mathbf{f}_{vu} + W_{tv}\mathbf{f}_u\right). \end{aligned}\qquad(8)$$

KAN enables a nonlinear fusion of atomic and bond-level features, allowing more expressive initial representations. Here W_{hv}, W_e and W_{tv} are weight matrices to scale the input feature into same dimension.

(2) KAN-enhanced attention mechanism

Instead of a fixed attention computation, KA-GAT learns adaptive attention scores via:

$$\begin{aligned} z_v^{(l+1)} &:= Wh_v^{(l)}, \quad z_u^{(l+1)} := Wh_u^{(l)},\\ \alpha_{vu} &:= \mathrm{Softmax}\left(h_{vu}^{(l)}\right), \quad m_{vu}^{(l+1)} := z_u^{(l+1)}\cdot\alpha_{vu},\\ m_v^{(l+1)} &:= z_v^{(l+1)} + \frac{1}{|N(v)|}\sum_{u\in N(v)} m_{vu}^{(l+1)}. \end{aligned}\qquad(9)$$

Here W is a value matrix. The use of KAN here enables adaptive, context-dependent attention scores, unlike standard GAT, which uses a fixed-weight attention mechanism.

(3) KAN-based feature update

After computing attention-based messages, we apply KAN for feature refinement:

$$h_v^{(l+1)} := \mathrm{KAN}\left(m_v^{(l+1)}\right), \quad h_{vu}^{(l+1)} := \mathrm{KAN}\left(h_{vu}^{(l)} + h_u^{(l)} + h_v^{(l)}\right).\qquad(10)$$

This ensures that node and edge features are updated in a nonlinear, data-driven manner.

(4) Readout and final prediction

Similar to KA-GCN, KA-GAT uses KAN-based graph-level feature aggregation, allowing a flexible representation of the molecular graph.

主要差异和贡献

KA-GNN 将 KAN 模块无缝集成到 GNN 的三个核心基本组成部分中，以显著提高模型的准确性和可解释性。

- 在所有三个核心组件上实现端到端的KAN集成：我们将KAN模块系统地嵌入整个GNN管道中——包括节点嵌入、消息传递和读出，从而构建一个统一的、完全可微分的架构，并增强学习能力。

- 基于傅里叶的KAN公式：我们引入了一种基于傅里叶级数的KAN变体，取代传统的基于B样条或RBF的方法。我们的傅里叶-KAN提供了灵活的函数逼近能力，使模型能够捕捉节点特征和注意力权重中的平滑与高度非线性模式，同时具备可证明的泛函逼近能力。

- 自适应特征转换与注意力建模：在 KA-GAT 中，我们用 KAN 模块替代了传统的基于 MLP 的注意力机制，使得对共价和非共价相互作用的建模更加灵活。KA-GCN 和 KA-GAT 都受益于更丰富的非线性转换能力，其表现力超过了 ReLU-MLP。

- 可解释性与领域对齐：除了精度提升外，KA-GNN还提供了更好的可解释性。如图3所示，该模型能够持续突出功能上重要的子图，提供具有化学意义的解释，其与专家策划知识的对齐优于基线GNN。

数据可用性

本研究使用的所有数据可通过 GitHub 获取，网址为 https://github.com/LongLee220/KA-GNN，或通过 Zenodo 获取，网址为 https://doi.org/10.5281/zenodo.15003782（参考文献 58），也可以从原始数据集存储库下载，网址为 https://moleculenet.org/datasets-1。原始数据随本文提供。

代码可用性

所有代码可通过 GitHub 获取，网址为 https://github.com/LongLee220/KA-GNN，亦可通过 Zenodo 获取，网址为 https://doi.org/10.5281/zenodo.15003782（参见文献 58）。

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Key differences and contributions

KA-GNN seamlessly integrate KAN modules into the three fundamental components of GNNs, to notably enhance the model accuracy and interpretability.

- End-to-end KAN integration at all three core components: we systematically embed KAN modules into the entire GNN pipeline-including node embedding, message passing and readout, resulting in a unified, fully differentiable architecture with enhanced learning capacity.
- Fourier-based KAN formulation: we introduce a KAN variant based on the Fourier series, replacing traditional B-spline or RBF-based approaches. Our Fourier-KAN offers flexible function approximation, enabling the model to capture both smooth and highly nonlinear patterns in node features and attention weights, with provable universal approximation capabilities.
- Adaptive feature transformation and attention modelling: in KA-GAT, we replace conventional MLP-based attention mechanisms, with KAN modules, enabling more flexible modelling of both covalent and non-covalent interactions. Both KA-GCN and KA-GAT benefit from richer nonlinear transformation capabilities, surpassing ReLU-MLPs in expressiveness.
- Interpretability and domain alignment: in addition to accuracy gains, KA-GNN offers improved interpretability. As illustrated in Fig. 3, the model consistently highlights functionally important subgraphs, providing chemically meaningful explanations that outperform baseline GNNs in aligning with expert-curated knowledge.

Data availability

All data used in this study are available via GitHub at https://github.com/LongLee220/KA-GNN and via Zenodo at https://doi.org/10.5281/zenodo.15003782 (ref. 58) or can be downloaded from the original dataset repository at https://moleculenet.org/datasets-1. Source data are provided with this paper.

Code availability

All code is available via GitHub at https://github.com/LongLee220/KA-GNN and via Zenodo at https://doi.org/10.5281/zenodo.15003782 (ref. 58).

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Author contributions

K.X. conceived and designed the study. L.L. performed the calculations. G.W. provided computational resources. Y.Z. carried out the theoretical analysis. L.L., Y.Z. and K.X. jointly contributed to writing and revising the paper.

Competing interests

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